

Cross-Modal Transfer Learning for Solar Panel Defect Detection

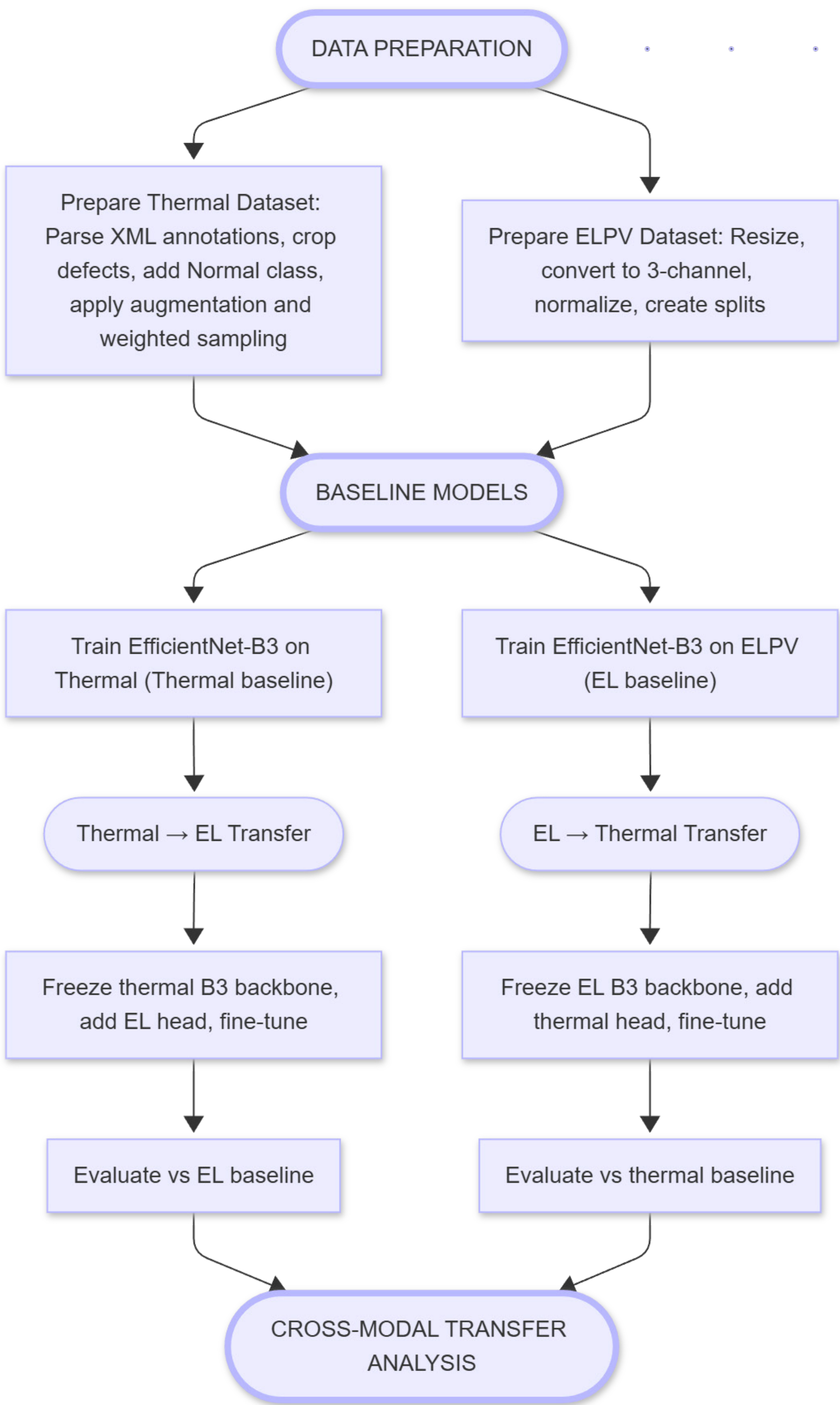
Introduction

Solar panels develop defects such as hotspots, microcracks, broken cells, and inactive regions that reduce power output and increase maintenance costs. In industry, two different imaging modalities are used to detect these defects: thermal infrared imaging, which captures heat signatures during field inspection, and electro luminescence (EL) imaging, which reveals fine structural damage during factory testing. Because these modalities differ significantly in appearance and underlying physical information, manufacturers must maintain separate AI systems for each. This project investigates whether a cross-modal transfer learning approach can reduce this redundancy by allowing knowledge learned from one modality to improve defect detection in the other.

Observations

Model	Thermal Baseline (F1-Score)	EL Baseline (F1-Score)	EL → Thermal Transfer (F1-Score)	Thermal → EL Transfer (F1-Score)
EfficientNetB0	95.57%	89.5%	94.10%	88.40%
EfficientNetB1	95.39%	90%	95%	88.70%
EfficientNetB3	95.98%	91.7%	95.65%	89.10%

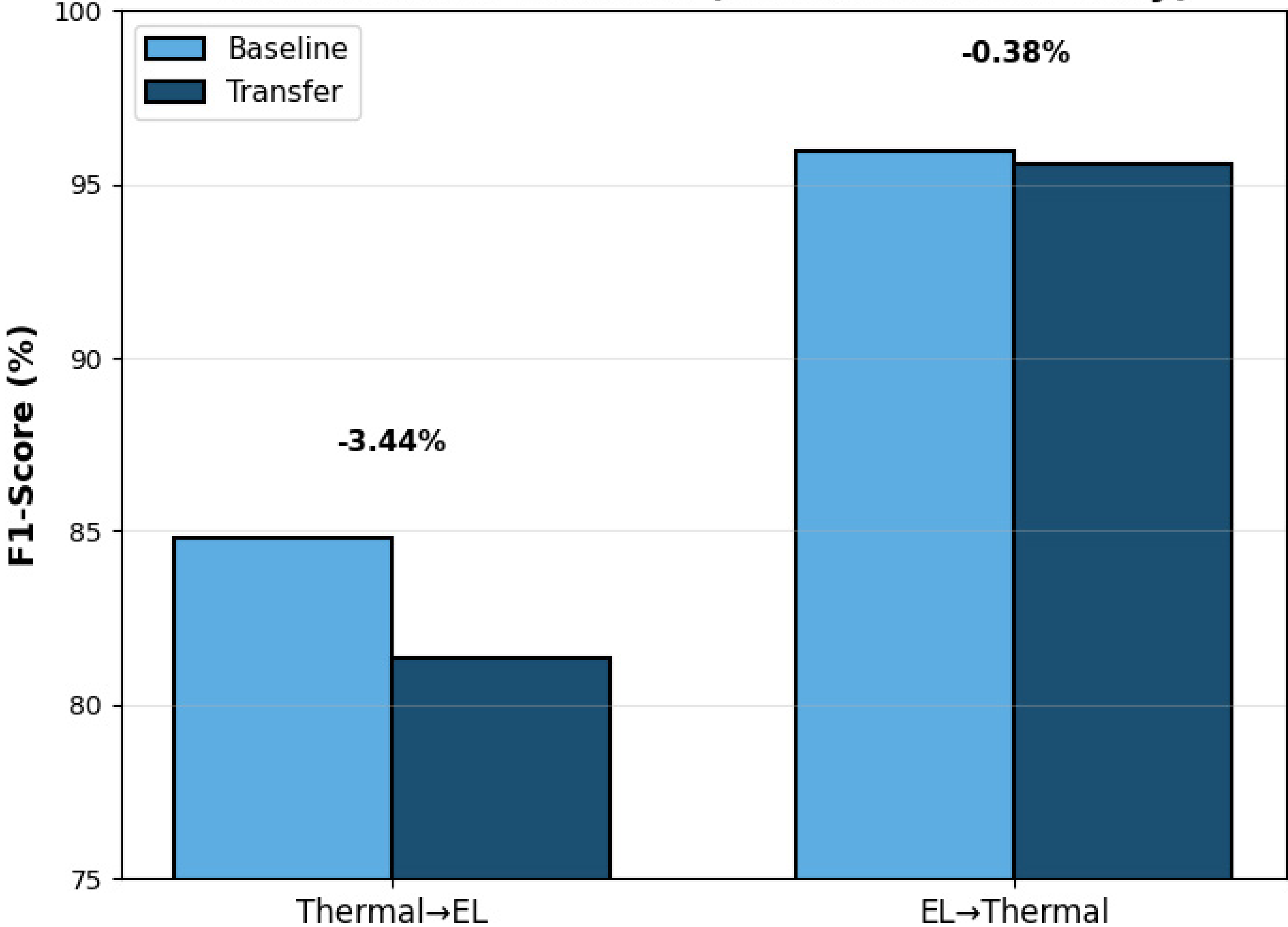
Methodology



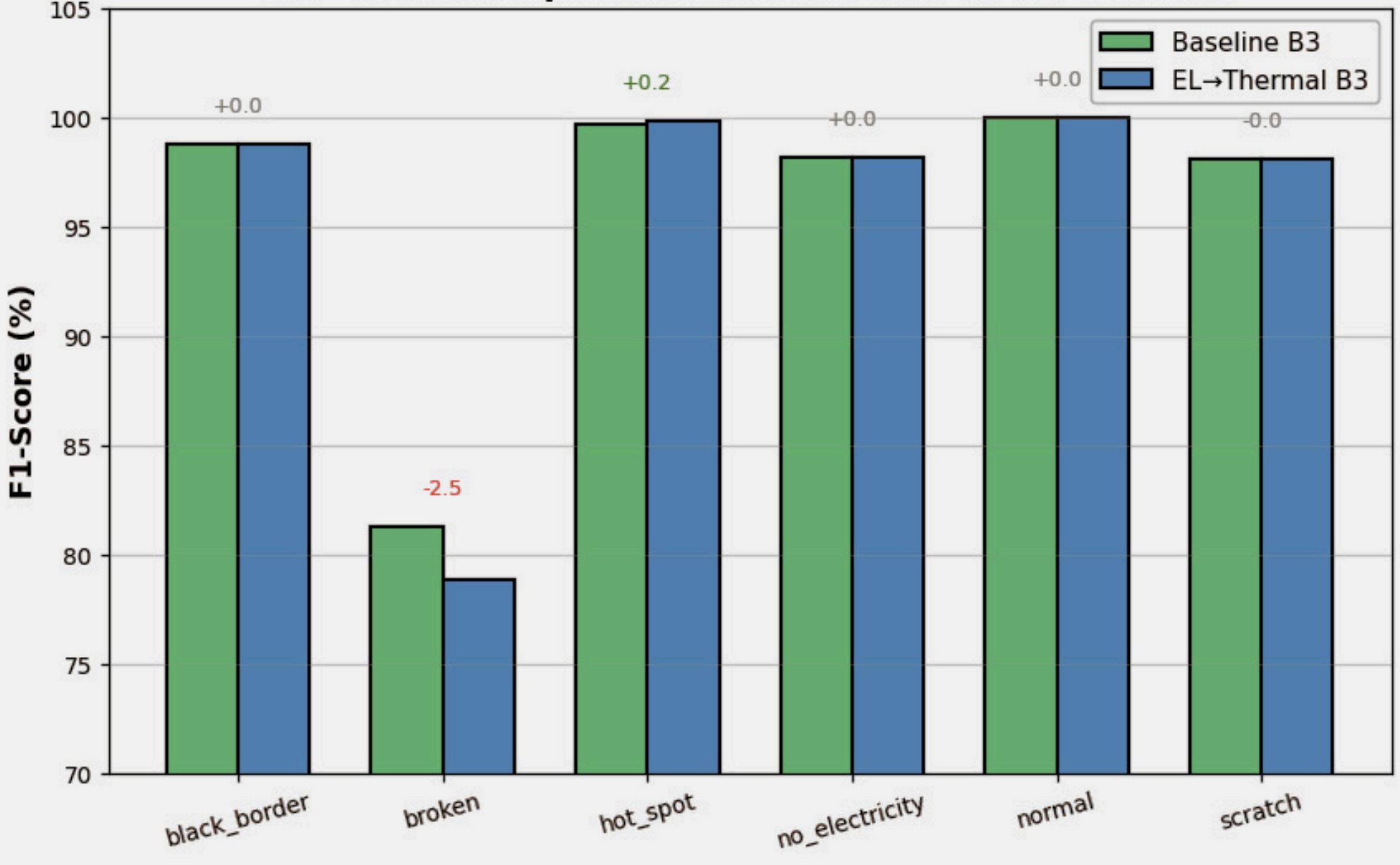
Key Findings

- Transfer learning is directional: EL to Thermal transfers well, while Thermal to EL is more limited.
- This arises from modality characteristics: EL contains fine structural detail, whereas thermal images capture broader heat signatures.
- The asymmetry highlights complementarity: each modality contributes unique information that can be combined for stronger inspection systems.

Bidirectional Transfer (EfficientNet-B3 Only)



Per-Class Comparison: B3 Baseline vs B3 Transfer



- Most defect classes (black_border, normal, scratch, no_electricity) show no meaningful change, indicating strong cross-modal generalization.
- Hot_spot improves slightly (+0.2%), suggesting EL structural cues help refine thermal predictions.
- Only broken decreases (−2.5%), reflecting modality mismatch - cracks visible in EL do not always appear in thermal images.
- Overall, EL→Thermal transfer successfully preserves class-level performance across 83% of classes (5/6 unaffected), confirming that EL features are broadly compatible with thermal defect detection.

Future Work

A promising next step is multimodal fusion, where thermal and EL images are combined within a single model to leverage the complementary strengths of both modalities. EL provides high-resolution structural detail, while thermal imaging captures operational heat patterns, and fusing these signals could enable more comprehensive and accurate defect detection than either modality alone.